

Title

MoBI-View: An Open-Source Realtime Biosignal Visualization Tool for LSL

Authors

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Keywords (2-5)

MoBI, multimodal data, data visualization, LSL, quality control

Category

Talk, Non-Experimental Work

Abstract**Introduction**

Mobile Brain/Body Imaging (MoBI) is an integrative multimodal approach for studying neural activity, motor behavior, and physiological processes during naturalistic, embodied behavior (Gramann, 2019; Makeig et al., 2009). By combining brain imaging with measurements of body movement and environmental interaction, MoBI enables investigation of cognition during active behavior rather than constrained laboratory tasks. Such experiments typically involve simultaneous acquisition of electroencephalography (EEG), motion capture, eye tracking, physiological signals, and audiovisual data. Preserving temporal relationships among these modalities is fundamental for meaningful interpretation of brain–behavior coupling (Jungnickel et al., 2019).

Lab Streaming Layer (LSL) is an open-source networked middleware that provides a framework for time synchronized multimodal data acquisition (Kothe, 2014). While LSL has been widely adopted amongst researchers, challenges remain in monitoring data integrity and signal quality during live data collection. Existing real-time visualization tools such as `Stream Viewer` (Intheon, 2016) and `mobilab` (SCCN, 2017) often rely on proprietary platforms or unmaintained legacy frameworks. Unlike MoBI-View, these tools typically function as monolithic desktop applications requiring local execution. This architecture creates compatibility barriers with modern Python-centric workflows and hinders distributed monitoring, making it difficult to deploy monitoring solutions on auxiliary devices for immediate experimental intervention. To address these challenges, we present MoBI-View, a lightweight, web-based platform for real-time visualization of synchronized multimodal data streams during MoBI experiments.

Methods

MoBI-View uses a Python-based modular Model–View–Presenter architecture. LSL streams are handled through a unified `DataInlet` abstraction, which manages stream discovery, efficient circular buffering, synchronization, and error handling across modalities. A central Presenter orchestrates data flow between acquisition and

visualization, and performs lightweight preprocessing such as configurable bandpass filtering, utilizing the circular buffers for continuous time-domain filtering. This decoupled design allows extensive quality control without refactoring core components.

The browser-based View layer handles client-side interaction without disrupting data flow. Time-resolved data frames stream as structured JSON objects over `WebSockets`, supporting sub-second visualization stable across varying stream counts and browser types. This GUI free design allows headless operation on acquisition machines to conserve network bandwidth while enabling remote monitoring from standard web browsers.

Results

MoBI-View has been deployed in numerous completed multimodal data collection sessions across six sites (two in the US, two in South Africa, two in Brazil). It enabled stable real-time visualization of synchronized EEG, motion capture, eye-tracking, physiological, and audiovisual streams. Live monitoring supported early identification of connection interruptions, packet irregularities, and environmental disturbances, allowing immediate adjustments during acquisition rather than solely during post-hoc analysis.

Discussion

MoBI-View provides a flexible foundation for real-time multimodal data monitoring in naturalistic neuroscience experiments. By combining a modular backend architecture with a web-based visualization layer, the platform simplifies deployment, improves accessibility, and supports extensible preprocessing and quality control workflows. Ongoing development includes integration of automated motion-based classifiers and expanded quality control metrics (such as a live EEG impedance checker), supporting more reliable and efficient MoBI data collection as experimental complexity increases.

Word count: 477

Figure (For the Abstract)

Title: High-level architecture of MoBI-View

Caption: Multimodal LSL streams are synchronized and aggregated by a central presenter and streamed in real time via WebSockets to browser-based visualizations, enabling platform-independent monitoring during MoBI experiments.

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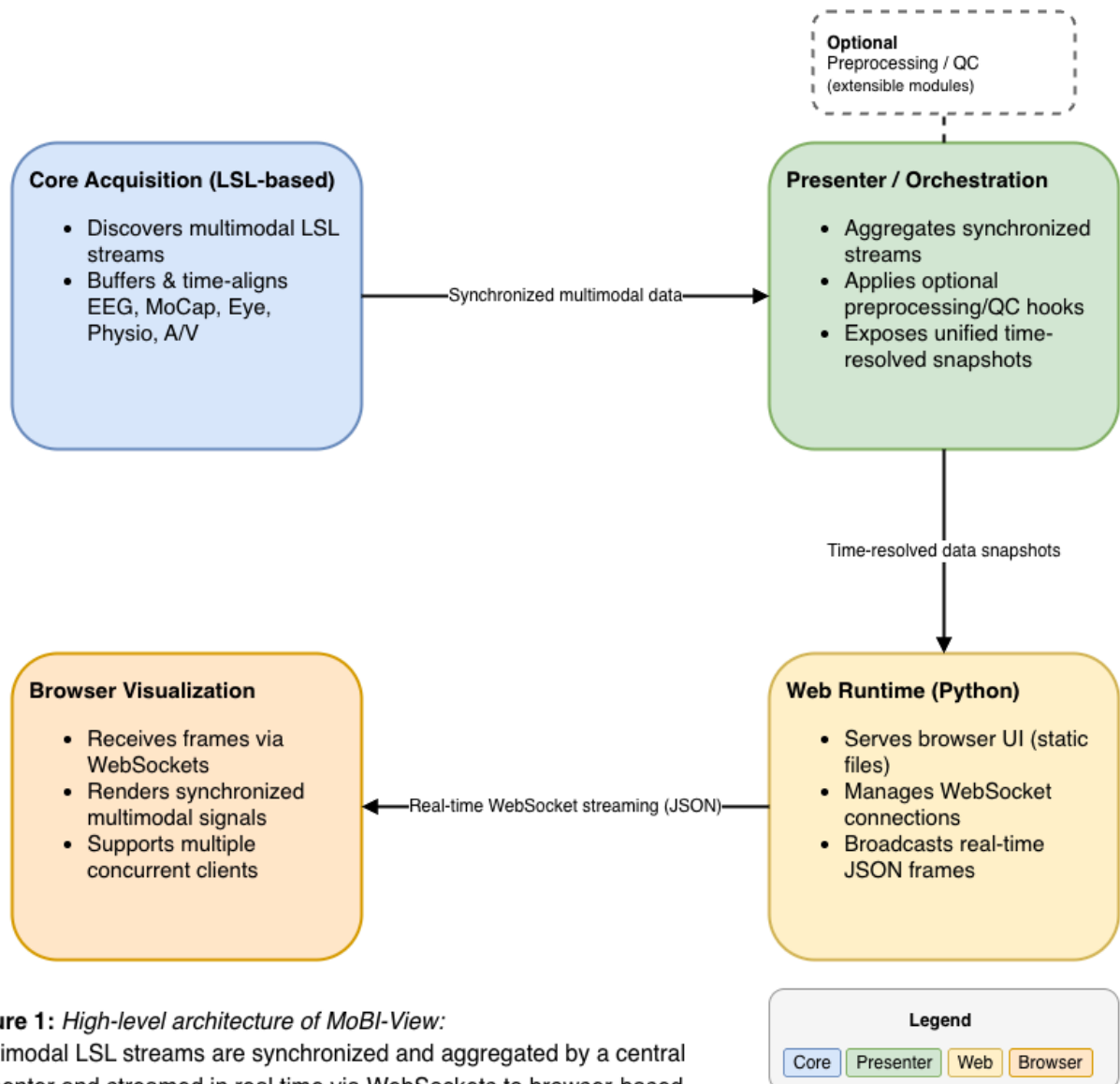


Figure 1: High-level architecture of MoBI-View:

Multimodal LSL streams are synchronized and aggregated by a central presenter and streamed in real time via WebSockets to browser-based visualizations, enabling platform-independent monitoring during MoBI experiments.

Figure (For the Poster)

Title: Unified Modeling Language (UML)-style schematic of the MoBI-View architecture.

Caption: The diagram highlights data flow between Model–Presenter components and a browser-based View, illustrating how LSL-based acquisition, real-time WebSocket broadcasting (web.broadcaster), visualization modules, and user-defined processing/QC routines interact within the system.

(https://drive.google.com/file/d/19L9tWqeiklg_NQaMBR-8Op1AzcPkEaF4/view?usp=sharing)

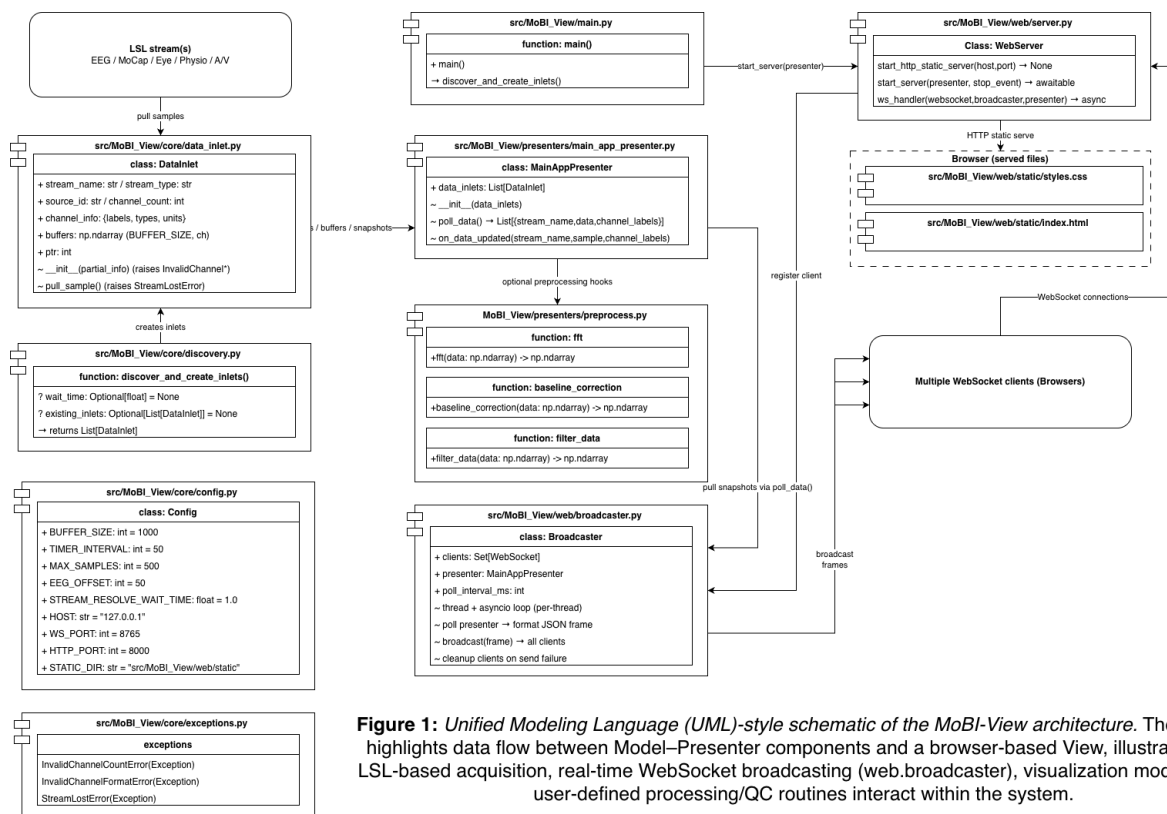


Figure 1: Unified Modeling Language (UML)-style schematic of the MoBI-View architecture. The diagram highlights data flow between Model–Presenter components and a browser-based View, illustrating how LSL-based acquisition, real-time WebSocket broadcasting (web.broadcaster), visualization modules, and user-defined processing/QC routines interact within the system.

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